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Sentiment analysis of financial news for stock price prediction: Empirical evidence from India

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Research Article

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Abstract

The stock prices are majorly affected by news related to macroeconomic conditions of the country as well as the global economy. The stocks are also affected by news specifically related to them. In this study we tried to understand how news affects the change in stock indices of emerging markets. We have used NIFTY50 and headlines of the Economic Times newspaper for this study. The model gives information on whether the NIFTY50 index will rise or fall given the news headlines. We have made three models based on Random Forest Classifier, Decision Trees Classifier and Multinomial NB Classifier. The Random Forest Classifier based achieved the best precision of 0.73. This model can give the investors greater decision support.

1. Introduction

In the current day and age, the stock market is one of the most important components of the global and local economies. The stock market is a secondary financial market where companies can issue stocks through a process known as initial public offerings (IPOs) and these stocks are then traded between retailers and institutions through buying and selling. Stock markets are associated with high rewards, but they also have dangers that might result in a loss. This makes stock market forecasting a fascinating but difficult task, as significant volatility, irregularity, and noise make it impossible to anticipate stock markets with high accuracy.

Several approaches and theories have been developed to aid traders in predicting market movements and explaining the stock market's growth and collapse.

Due to its dynamic and unpredictable nature such that even the best of traders/analysts cannot predict where a stock would move in the future, stock market prediction (SMP) is not an easy task. SMP is a type of time-series forecasting that evaluates historical price movements, detects a pattern and predicts future prices. Analysts from several fields, such as economics, mathematics, material science, and computer science, have been concerned with financial market forecasts.

The ability to generate money through stock trading is a key component in stock market forecasting. The price of a share, the performance of the firm, government regulations, the country's gross domestic product (GDP), inflation, natural catastrophes, and other variables all impact the stock market. According to the efficient market hypothesis (EMH), the markets are efficient, leaving no room to make excess profits by investing since everything is already fairly and accurately priced, this implies that the stock market values are significantly impacted by fresh information and follow a random walk pattern, making them hard to predict using past data alone. This was a widely accepted belief in the past. Researchers were able to demonstrate that stock market values might be forecasted to a degree, thanks to technological advancements. Stock market movements may be predicted using historical market data along with news headlines or social media headlines that may create a significant impact on the market the following day.

Previous papers have focused on using Machine Learning to conduct sentiment based analysis on the American stock market, with some success. We wanted to see if we could apply that work to the stock market in emerging market economies and see if sentiment based analysis can provide a high level of information to the relevant stakeholders.

2. Review Of Literature

Prediction of trend in stock price is an area of research which has been gaining traction for a long time now. The researchers have been trying to improve the accuracy as much as possible as higher accuracy means higher returns. As a result, in recent years a lot of effort has been put into building models which can foresee future price returns of specific stocks or the entire market. In the bulk of existing methodologies, technical indicators are utilised. Some of the studies observed that corporate news items and stock price fluctuations had a strong correlation. Extraction of data from sentences and texts has been the subject of several research.

An approach for creating phrase representations that employs a weighted average of task vectors has been proposed by Sanjeev, Yingyu Liang, and Tengyu. They use text continuity to train an encoder-decoder model, and then the vector generated by this process is used to represent the phrase. Another important usage aspect of their research was BERT. Sentence-BERT is a pre-trained BERT network that uses a siamese and triplet network architecture to create sentence structures which are logically meaningful. Le and Tomas worked on similar lines by creating a vector of paragraphs that enables the structure of texts of various lengths, such as papers and paragraphs.

Recent works in the topic of natural language-based financial forecasting were studied by Frank and Erik. This study looks at earlier research on the subject, which includes research from the early 1980s uptill the most recent breakthroughs on studies using data from news sources and social media. Many publications in the topic of using social media material focus on Twitter data because of its clear semantics and short duration. Machine learning methods have also been used extensively in this field. Xioa and Yue employ neural tensor networks and convolutional networks to predict the S&P 500 index based on news content.

In 2002, Antonina Kloptchenko, Tomas Eklund, Barbro Back, Jonas Karlsson, Hannu Vanharanta, and Ari Visa used data and text mining technologies to uncover hidden trends in financial reports' quantitative and qualitative features. Annual reports, they believe, are an important tool for companies to communicate with their stakeholders. Their text mining technology, on the other hand, can only be utilised on yearly reports, not on websites or news archives.

In 2007, Vatsal H. Shah and Dr. Mehryar Mohri emphasised the usage of Support Vector Machines when they presented machine learning strategies for stock prediction. They also advocated the use of Linear Regression, decision stump prediction, expert weighting, and online learning, as well as the benefits and drawbacks of each method. With the machine learning methodologies they outlined, they focused on text and language processing.

The research by Mattia Vicari and Mauro Gaspari analyses if and to what extent it is feasible to trade on news sentiment, and whether deep learning (DL) would be an appropriate technique for doing so. DL is designed specifically for dealing with large volumes of data and executing complicated tasks that need automated learning. It seeks to find complicated patterns in a dataset, which may appeal to investors who want to enhance their trading process. Furthermore, given its capacity to "remember" preceding words in a phrase, DL and especially LSTM appear to be a viable choice from a linguistic standpoint. They utilize this technology to estimate market sentiment using news headlines after explaining how DL models are formed. The forecast is based on the Dow Jones industrial average and is based on an analysis of 25 daily news stories from 2008 to

2016, with an extension to 2020. The outcome may then be utilized to construct an algorithmic trading strategy as an indication. They looked at two unique cases for testing that were followed across five time steps.

3. Data And Methodology

3.1. Data collection

We have aimed to conduct sentiment analysis on the stock market in India and decided to analyse NIFTY50 along with the news articles from economic times for the same time period. The paper primarily used python along with its scrappy module to web-crawl and scrape data from The Economic Times website from January 1st 2008 to March 18th 2022. The dataset contains all unique headlines of articles published by Economic times day wise on trading days. Alongside this, we have also collected daily transactional data of NIFTY50 including Opening Price, Closing Price and Transaction Volume and consolidated these 2 datasets for each trading day. We have created a matrix of this data day-wise and labelled each day in binary terms depending on whether NIFTY50 has gone up or gone down, This has been combined with the collection of news headlines on that particular day as shown below.

Table 1
Snapshot of data.

Date	*Label*	*Top1*	*Top2*	*Top3*	*Top4*	*Top5*	*Top6*
03-01-2008	0	A 'hindrance to operations': extracts from the leaked reports	Scorecard	Hughes' instant hit buoys Blues	Jack gets his skates on at ice-cold Alex	Chaos as Maracana builds up for United	Depleted Leicester prevail as Elliott spoils Everton's party
04-01-2008	0	Scorecard	The best lake scene	Leader: German sleaze inquiry	Cheerio, boyo	The main recommendations	Has Cubie killed fees?
05-01-2008	0	Coventry caught on counter by Flo	United's rivals on the road to Rio	Thatcher issues defence before trial by video	Police help Smith lay down the law at Everton	Tale of Trautmann bears two more retellings	England on the rack
06-01-2008	1	Pilgrim knows how to progress	Thatcher facing ban	McIlroy calls for Irish fighting spirit	Leicester bin stadium blueprint	United braced for Mexican wave	Auntie back in fashion, even if the dress looks a bit tatty
07-01-2008	1	Hitches and Horlocks	Beckham off but United survive	Breast cancer screening	Alan Parker	Guardian readers: are you all whingers?	Hollywood Beyond
10-01-2008	1	Fifth round draw	BBC unveils secret weapon in ratings war: South Melbourne	Second Division round-up	European round-up	Third Division round-up	Welfare could claim Killie tie as Caley Thistle land Celtic
11-01-2008	1	Man Utd 2 – 0 South Melbourne	How North Atlantic drift could carry away Old Firm	Buoyant BBC to show Brazil final live	Tranmere given all-clear in the Cup	United sit poorly with the Doc	Queen's Park peril clouds Hampden future
12-01-2008	0	Newcastle seek new football supremo	Liverpool aim to speed up Heskey deal	Highlanders voted up	Edwards' power play suffers new blow	Chelsea gamble on Weah	Taylor settles the eternal tie

Source: Authors' extraction of data

Date	*Label*	*Top1*	*Top2*	*Top3*	*Top4*	*Top5*	*Top6*
13-01-2008	1	Bungling officials on the carpet	And in the red and raw corner it's 'Killer' MacKenzie	United put their shirts on 30m	England against plan for home nations' revival	Donald poised to quit Test scene	Adams stares into the abyss
14-01-2008	1	Pompey plump for Pulis work ethic	Roma under fire over Rolexes for referees	Prenton Park Two told to take a break	OK, I didn't figure in Rio but I'm still No1 with the under-10s	Chelsea tune in to Weah's world	Top storey awaits the Cottage
18-01-2008	1	England's elite set for pay struggle	Robson puts his job on the line	Old Firm lobby to replace Uefa Cup	Babbel to check in at Anfield	For Muralitharan read MacGill	All padded up with nowhere to go
19-01-2008	0	No apologies from the captains for contrived finale	One-off deal sets up Hussain Test victory	Old Trafford in talks to reduce Yorke's national service	Draper sets out his stall in Madrid	Forest giant still standing	Adams stays on Arsenal injury roll

Source: Authors' extraction of data

The figure below depicts the steps followed in data processing and analysis.

3.2. Pre-processing

Data must be preprocessed before being fed into a machine learning model once it has been obtained. The relevance of the output is decided by the pre-processing of the data. For use in a machine learning model, it is important that the textual data is transformed into a proper structured format. There are three key preprocessing phases, according to previous research: feature selection, order reduction, and feature representation.

However prior to this, all news headlines of one particular day have to be clubbed into one statement. All special characters have to be removed and all statements have to be converted to lowercase for the sake of uniformity.

3.2.2. Feature Selection

In textual data processing, feature selection is critical. The majority of the research employed basic feature extraction approaches like Bag of Words, which breaks the text into words and then changes words into a numeric feature. The amount of occurrences of a term influences the feature selection. The same feature

selection procedure was employed in this study. However, there is one drawback to this method: the order of words is lost, resulting in a loss of context.

3.2.2. Order Reduction

The number of features increases as a result of the feature selection procedure for textual data. High-dimensional features are particularly difficult to process, which causes most learning algorithms to be inefficient. This is called the curse of dimensionality. The training complexity of the algorithms will be reduced as the number of features decreases. Principal Component Analysis (PCA), a well-known type of multivariate analysis, is used to choose the most important data while minimising the dimensionality of the features.

3.2.3. Feature Representation

One of the most significant components in the training of machine learning algorithms is “feature representation”. The input data is transformed to a numeric form of representation which can now be easily processed by machine learning algorithms once the number of required features has been calculated.

For the bag of words, boolean representation is a feature representation technique in which the existence and in absence of a feature (word) are represented by 1 and 0, respectively. This is called count-vectorization. It convertsCount-vectorization each statement into a vector of 0s and 1s.

3.3. Classifier learning

We studied a lot of machine learning models which can be used for sentiment analysis and stock price prediction. A few of them have been summarised below followed by the details of the models which we have used for this study.

3.3.1. Artificial Neural Networks (ANN)

The ANN is a biologically inspired technique for solving complicated problems that uses a large number of artificial neurons that are tightly coupled. These models provide simpler representations of an issue by performing several transformations on the feature space, followed by non-linearity. For sentiment analysis, ANN models have been used in a number of research. For example, the authors used ANN to forecast the S&P 500 index's daily trajectory. The dataset was streamlined using three-dimensional reduction approaches such as Kernel-based Principal Component Analysis, Fuzzy Robust Principal Component Analysis (FRPCA), and Principal Component Analysis. The findings indicated that combining ANNs and PCAs is more efficient. Also, the choice of a suitable kernel function has a direct impact on KPCA's performance.

For NASDAQ price prediction, researchers tested three ANN models: MLP, dynamic artificial neural network, and GARCH model. The Mean Absolute Deviation (MAD) and Mean Square Error were used to analyse all three models (MSE). The MLP outperformed DAN2 and GARCH-MLP, according to the findings.

3.3.2. Support Vector Machine (SVM)

The SVM is a method which involves supervised machine learning that reduces error. It also improves geometric margins. It basically is a method for pattern classification and regression. In terms of accuracy, the SVM linear separation technique, which is significantly superior when compared to other classifiers.

3.3.3. Naïve Bayes (NB)

Naïve Bayes is a classification approach that uses the Bayesian Theorem of Probability to classify data points. This classification approach is very quick and can handle big datasets.

For example, for sentiment analysis on textual data from multiple sources, the authors used the Naive Bayes algorithm. The authors analysed the impact of traditional and social media data sources on various businesses and their interconnectedness.

3.3.4. Deep Neural Networks (DNN)

DNNs use a lot more hidden layers and neurons to extract and modify features on their own which makes it a lot more advanced. Higher the number of hidden layers with non-linear processing units higher will be the efficiency of learning from raw input. DNNs are commonly used to produce financial projections using textual and numerical data.

For example, a recent study examined the Auto-Regressive Integrated Moving Average (ARIMA), Vector AutoRegression (VAR), LSTM, and Nonlinear Auto-Regressive with exogenous inputs to four stock market price prediction models. The model's performance was evaluated using accuracy as a metric.

The NASDAQ closing price was used as the foundation for the analysis in that study. NARX made accurate short-term forecasts, but failed to make long-term projections, according to the research. Models that integrated machine learning and technical indicators were also discovered to provide more accurate predictions. LSTM networks can learn long-term dependencies, giving them a strong effect on time series prediction.

The authors of the same study also made a Nifty price prediction model which employs dynamic LSTM and uses Open, High, Low and Close as features. This model gave a

Root Mean Square Error (RMSE) of 0.00859 in daily percentage changes.

3.3.5. Regression Algorithms (RA)

The relationship between a dependent variable and its independent factors can be established using a method college Regression. Simple linear regression, multiple regression, decision tree regression, logistic regression, support vector regression (SVR), and ensemble regression have all been employed in earlier investigations. The authors for example, built a model that forecasts the stock price of a business specified by the user a few days in advance. The data, which came from a variety of sources, was subjected to regression analysis and candlestick pattern detection. The model anticipated that the market would reach an acceptable level of efficiency. Also, various machine learning techniques were applied, resulting in an 85 percent increase in accuracy.

3.3.6. Tree-Based Algorithms

In tree-based models, a decision tree is used to depict how different input variables might be utilised to predict a target value. Machine learning uses tree-based models for both classification and regression problems. The input variables are repeatedly subdivided in order to build the decision tree, and each branch is evaluated for prediction accuracy as well as efficiency and effectiveness. By separating the variables in a different order, the

number of layers and computations necessary to make an accurate forecast can be lowered. The most relevant factors (those that have the largest effect on the prediction) are placed at the top of the tree hierarchy, while unimportant qualities are eliminated when developing a successful decision tree.

For this study we have used three different models namely RandomForest classifier, Decision Trees classifier and Multinomial NB. The first two models are based on Tree-Based algorithms and the last is based on Naïve Bayes algorithm.

4. System Evaluation

For stock market forecasting, two methodologies are commonly used: classification and regression. The first method(categorization or classification) divides the market movement into 2 group. This grouping is conducted by the use of binary numbers that are assigned to each group so as to provide a numerical value so that prediction can be done accordingly.

The bulk of prior studies employed accuracy as a criterion, which is defined as the proportion of right predictions divided by the total number of test instances. The MSE (Mean Squared Error) is a measure that determines the average difference between expected and actual output. The ratio of properly predicted positives to total positives expected is known as precision. Recall is defined as the proportion of properly predicted positives to the total number of positives. The F-score is the harmonic mean of accuracy and recall, and it reveals how important false positives and negatives are.

5. Evaluation

We have chosen to use 3 algorithms and compare their results holistically in order to conduct the study namely:

1) Multinomial Naive Bayes

2) Random Forest Classifier

3) Decision Tree Classifier

Table 2
Accuracy depiction

	Accuracy (Weighted Average)	*f1-score*
RandomForest Classifier	0.73	0.69
MultinomialNB	0.71	0.49
Decision Trees	0.58	0.58

On basis of these results, we can conclude that the MultinomialNB and RandomForest Classifier produce more accurate results, while their weighted averages are similar, the RandomForest Classifier has a higher F1-Score and therefore would be the preferred classifier algorithm as confirmed by its use in References for sentiment analysis for American Stock Market data.

5.1. Overfitting:

Overfitting is one of the most difficult problems in machine learning models wherein the model adapts itself according to the training data and therefore does not produce any tangible results on the test data or data from any other set. This implies the model detects noise or random oscillations in the training data and learns them as concepts. These concepts are inapplicable to the new data that has to be forecasted, resulting in poor model generalisation. Because stock market data is very erratic, it's critical to describe how this problem was solved. Regularisation is the most significant safeguard against overfitting. The superfluous weights from the chosen features are removed and redistributed equally using this procedure.It discourages the learning of complicated or flexible models, hence reducing the danger of overfitting. We accomplished this by deleting unnecessary weights from our bag of words that may apply to the time period of our training data but not to the time period of our testing data; another method we tackled such difficulties was to use alternating years as training and testing data, reducing the time period component.

6. Conclusion

For investors and traders, financial markets provide a tremendous platform, since they may trade from any device with an internet connection. As with many other aspects of life, technological improvements have changed the stock market. It is now possible for people to raise the value of their possessions. The way people trade stocks has only changed as a result of internet trading. The budgetary markets have evolved rapidly, transforming into a globally interconnected economy. Such advances in the global markets has allowed new opportunities in the field of machine learning and big data analytics that would allow for better decision-making by analysts with respect to the movement of the market.

We examined literature using a broad SMP framework, with a focus on studies published in the last decade (2011–2021) while conducting our own research, taking inferences on the type of data utilised as input, data pre-processing techniques, and machine learning approaches used to produce predictions. We also looked at how different research used different evaluation measures to gauge performance. A thorough comparative study was also undertaken, and it was discovered that the Random Forest Classifier is the most widely used SMP technique.

We could initially conclude with a 50% probability that 1 particular news article would cause the market to go up or down on a particular day, this probability has turned out to be more than 70% after running and comparing results from the 3 classification algorithms that we have used, therefore we can now safely conclude that traders could use news-based/sentiment-based trading as one of the parameters in their strategies and that it would likely increase their chances of being correct.

Furthermore, this study can be used as a basis for another study in order to find out periods during a year(Budget, Monetary Policy Meetings) in which the market might be more heavily reliant on sentiment based study(more than 72%), this would help traders in understanding the impact of such events and reducing their reliance on technical indicators in such periods/ keeping more conservative stop loss for their trades, it would also help the Regulatory bodies in deciding the timings for such events so as to not cause a major shake-up in the market.

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Figures

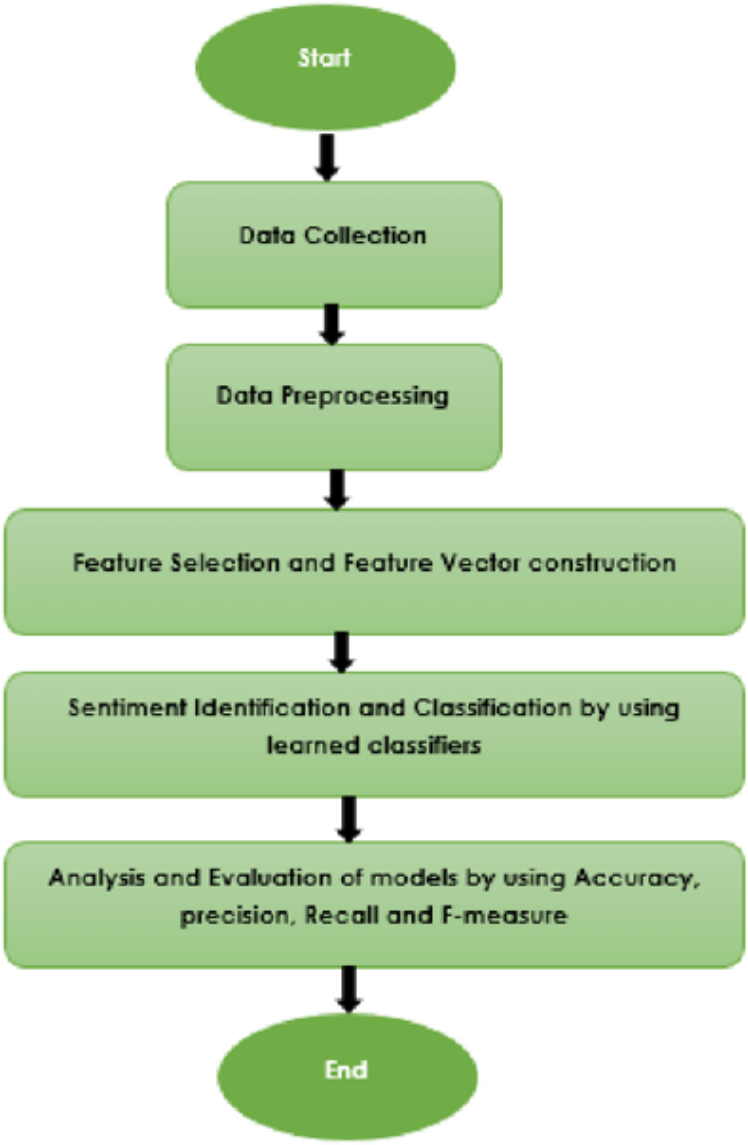


Figure 1

Steps in Data Analysis